Word embeddings quantify 100 years of gender and ethnic stereotypes
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Jayaram Gokulan – jg929 - Critique #3

Time is a powerful construct; from the elements of Darwinism to the latent transformations of the modern era, it is perhaps the single most significant driver of change that deserves just attention. The evolution of ethnic stereotypes and gender roles are classic demonstrations of the same. As a noted phenomenon of the 1900s that lost its steam towards the near end of the millennium, its transformational trajectory is fascinating and one that the paper has presented well in the context of word embeddings techniques. Juxtaposing word associations against the reformations of a social landscape reminds the reader of the versatility of such methods but prompts the larger question of the actual influence of temporal dynamics in general; are they truly limited to the more familiar artifacts of life alone? Admittedly, an overlooked impact of temporal mutations is that on the literary elements themselves than the objects they seek to describe. Indeed, over time, words, phrases, and even usages have undergone dramatic changes to represent entirely different senses than originally intended. For instance, while semantic progressions of adjectives such as 'awful, 'egregious,' or 'silly' saw a reversion, to receive its negative undertones that we know today, that of a word like 'nice' went on to bear a more benevolent meaning. The implications are then apparent; would the appearance of such word association bias between genders or ethnic groups, through the progression of time (historical and contemporary), necessarily suggest a semantic, thus sociological level change or status quo? A conclusive answer may be out of the question. However, any analysis forgoing such information could potentially invite some form of contradiction and inconsistencies to their final results.

While the works on dynamic word embeddings (Bamler and Mandt, 2017) offer related insight into the above discussion, another rationale to consider, in the light of similar deliberations, could be contextual word embeddings. Contextual level information strongly influences the semantics of words and hence their interpretation and representation. Then, it stands to reason that the sources referenced in the study for verifying embedding relationships need not be an exception to this principle. As a case in point, while a word such as 'accounting' usually connotes to its occupational sense of semantics, it can also find its place as a part of a phrasal verb (in the usage 'accounting for') within the same context. The same could be inferred for adjectives such as 'kind', which is perceived differently when used as a noun phrase (as 'kind of'). Considering how this might affect the bias measure for either genders or any ethnic stereotype, a notion to reconsider would be the conclusions drawn within this study, seeing how it may be based on possibly inaccurate metrics.

On a minor note, perhaps of interest to the sociology audience would be gauging the strength and extent of word associations outside the frames of past and present events. Ideas and evidence of possibilities before us often find a place in textual documents, such as journals and research papers, well ahead of their actual occurrence. Using similar articles to compare the word embedding structure of an entity of interest with base model embedding metrics (similar entities with the implications of an unforeseen event) could then offer insights into the unfolding of a future event for that entity. A simple analogy to this could be how contrasting the word embeddings of a deteriorating business, mentioned in a magazine, against that of historical companies with similar progression patterns, could convey a lot about the business as a going concern in the future.

Works Cited

Bamler, R. and Mandt, S. (2017): Dynamic Word Embeddings. arXiv:1702.08359 [stat.ML] (2017) (available at https://arxiv.org/abs/1702.08359).